Complete Preprocessing & Multimodal Fusion Pipeline

Detailed Technical Flow: Signal Conditioning, ECG-Guided Physiological Gating, Time-Frequency Transformations, and Hierarchical Multimodal Cross-Attention Fusion

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1. High-Level Architectural Pipeline

The automated screening pipeline ingests synchronous 30-second phonocardiogram (PCG, acoustic) and electrocardiogram (ECG, electrical) recordings acquired via an **EKO DUO ECG+ digital stethoscope** across four standard clinical auscultation positions: **APEX (Mitral)**, **LLSB (Tricuspid)**, **LUSB (Pulmonary)**, and **RUSB (Aortic)**. It detects **Heart Failure with Reduced Ejection Fraction (HFrEF, LVEF ≤ 40%)**.

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[Raw Stethoscope Signals (30s Synchronous ECG + PCG across 4 Auscultation Sites)]
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  ■■■ 1. Continuous Signal Preprocessing (Resampling + Zero-Phase Butterworth + Mains Notch Filter)
  ■■■ 2. Physiological Gating (Pan-Tompkins R-Peak Detection -> Anchored [t_R - 0.25s, t_R + 1.25s] S
lices)
  ■■■ 3. Feature Representations (PCG -> 32-Band Log-Mel 2D Spectrogram; ECG -> Normalized 1D Vector)
  ■■■ 4. Dual-Branch Encoders (PCG 2D ConvNet + ECG 1D Conv-Residual Encoders -> Latent Sequences)
  ■■■ 5. Level 1 Fusion: Inter-Modality Cross-Attention (Q_ECG x K/V_PCG -> Detects EMAT Timing Delay
)
  ■■■ 6. Level 2 Fusion: Inter-Site Spatial Fusion (Multi-Head Self-Attention across APEX, LLSB, LUSB
, RUSB)
  ■■■ 7. Feature-Decision Double-Level Multi-Heads (6 Simultaneous Supervisions -> Ensembled Predicti
on)
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2. Stage-by-Stage Detailed Preprocessing

Stage 1: Continuous Full-Recording Filtering & Caching

Raw 30-second recordings are conditioned as complete continuous waveforms prior to any window extraction:

- **Acoustic PCG Stream:**
  - **Polyphase Rational Resampling:** Resampled to **1440 Hz** using polyphase filtering (scipy.signal.resample_poly) to avoid phase artifacts.
  - **Zero-Phase Butterworth Bandpass:** 5th-order filter passing **20 Hz to 400 Hz** using forward-backward filtering (sosfiltfilt). This removes low-frequency respiratory motion (<20 Hz) and ambient room hiss (>400 Hz) with strictly zero phase distortion.
- **Electrical ECG Stream:**
  - **Sampling Rate:** Maintained at **500 Hz** (standard clinical monitoring resolution).
  - **Mains Notch Filter:** IIR notch filter at **50 Hz** (Q = 30) eliminates electrical powerline hum.
  - **Baseline Wander Removal:** 3rd-order Butterworth bandpass filter (0.5 Hz to 40 Hz) removes respiration baseline drift for clean R-peak detection.
- **Flat Memory-Mapped Cache:** Signals are stored sequentially in pcg.npy and ecg.npy. Metadata records per-recording coordinates (pcg_offset, ecg_offset), enabling sub-millisecond retrieval.

Stage 2: ECG-Guided Physiological Gating

Instead of arbitrary sliding windows that slice heart sounds mid-cycle, the synchronized ECG serves as a physiological clock:

- **Pan-Tompkins R-Peak Detection:** Evaluates 500 Hz lead-I ECG via derivative and moving-average integration to identify ventricular depolarization peaks (t_R).
- **Anchored Cardiac Cycle Windowing:** Extracts windows from **$[t_R$ - 0.25 s, t_R + 1.25 s]** (fixed duration: **1.50 seconds**).
  - **PCG Slice:** Exactly **2,160 samples** (1.50 s × 1440 Hz).
  - **ECG Slice:** Exactly **750 samples** (1.50 s × 500 Hz).
- **Acoustic Quality Verification:** Evaluates the Hilbert envelope to confirm S1/S2 heart sound peaks; rejects low-RMS (<0.005) or clipped microphone friction artifacts.

Stage 3: Window Feature Representation per Site

Each valid cardiac cycle window is converted into structured feature tensors for deep learning:

- **PCG Stream → 32-Band Log-Mel 2D Spectrogram:**
  - **Short-Time Fourier Transform (STFT):** 128-point Hann window with 64-sample hop (50% overlap).
  - **Mel Filterbank:** 32 triangular filterbanks spanning **20 Hz to 700 Hz**.
  - **Logarithmic Compression:** Log-Mel = log(max(Mel Power, 1e-6)) followed by standardization.
  - **Output Shape: (1, 32, 35)** — 1 channel, 32 frequency bins, 35 time frames.
- **ECG Stream → 1D Normalized Temporal Signal:**
  - Standardized: (x - mean) / std → **Output Shape: (1, 750)**.

3. Deep Multimodal & Multi-Site Fusion Mechanics

Level 1: Inter-Modality Cross-Attention (CCF / EMAT Detection)

In HFrEF, ventricular dyssynchrony prolongs the **Electromechanical Activation Time (EMAT)** — the latency between electrical excitation (Q-wave) and mechanical mitral valve closure (S1). The network discovers this latency through multi-head cross-attention:

- **PCG 2D Conv Encoder:** 3 Conv2d stages (32 → 64 → 128 channels) → sequence (B, 8, 128).
- **ECG 1D Conv-Residual Encoder:** 3 Conv1d stages (kernels 15, 7, 5; channels 32 → 64 → 128) → sequence (B, 8, 128).
- **Bidirectional Cross-Attention:** Q_ECG × K/V_PCG and Q_PCG × K/V_ECG adaptively query the timing between electrical spikes and acoustic heart sounds.
- **Fused Site Embedding:** Projected and pooled to a 128-D vector: $v_{site} \in \blacksquare^{128}$ (for APEX, LLSB, LUSB, RUSB).

Level 2: 4-Site Auscultation Spatial Fusion (CSB)

Because recordings from the 4 auscultation sites are collected sequentially, raw waveforms are asynchronous and cannot be concatenated in time. The network fuses them at the semantic embedding level:

- Stack the 4 site embeddings: [v_APEX, v_LLSB, v_LUSB, v_RUSB] → (B, 4, 128).
- **Inter-Site Multi-Head Self-Attention:** Models cross-valve correlations across the chest surface.
- **Noise Suppression:** If one site (e.g. APEX) contains local chest wall rubbing artifacts, the other 3 clean sites cross-reference and **suppress false alarms**.
- Produces the Channel-Shared Embedding: $v_{CS} \in \blacksquare^{128}$.

Level 3: Feature-Decision Double-Level Multi-Branch Heads

The architecture applies deep multi-task auxiliary supervision across 6 distinct heads:

- 4 Site Heads (P_APEX, P_LLSB, P_LUSB, P_RUSB) + 1 Shared Head (P_CS) + 1 Joint Multi-Branch Head (P_joint).
- **Total Training Loss:** $\blacksquare_{total} = \sum \blacksquare_{site} + \blacksquare_{CS} + \blacksquare_{joint}$ (with square-root class weights).
- **Final Decision Ensemble:** P_final = (P_APEX + P_LLSB + P_LUSB + P_RUSB + P_CS + P_joint) / 6.0.

4. Method Comparison Matrix

Model Architecture	Input Signals	Acoustic Feature	Cross-Attn	Multi-Site	AUROC	Recall	Accuracy
E1 Baseline	PCG Only	Haar DWT (cA2)	No	Post-hoc	0.584	88.3%	32.4%
E2 Baseline	PCG (Gated)	Haar DWT (cA2)	No	Post-hoc	0.581	73.3%	36.4%
E2 Enhanced	PCG (Gated)	Haar DWT (cA2)	No	Post-hoc	0.627	43.3%	56.6%
E3 Multimodal Fusion	ECG + PCG	Haar DWT + 1D ECG	No (Concat)	Post-hoc	0.618	80.0%	36.6%
Log-Mel 2D CNN	PCG (Gated)	32-Band Log-Mel 2D	No	Post-hoc	0.780	56.7%	87.2%
Hierarchical Cross-Attn	**4-Site ECG+PCG**	**Log-Mel + 1D ECG**	**Yes (EMAT)**	**Self-Attn**	**0.894**	**85.0%**	**85.2%**

Key Takeaway: At $\tau = 0.23$, the Hierarchical Multimodal Cross-Attention network achieved **85.0% Recall and 85.2% Accuracy** simultaneously on the in-house cohort (AUROC 0.894), and **100.0% Sensitivity with 0.990 AUROC** on the independent 120-subject community scenario test set.